Some useful Gaussian Integrals

1) Calculate $I_0(\alpha) = \int_{-\infty}^{\infty} du \; e^{-\alpha u^2/2} = \sqrt{\frac{2\pi}{\alpha}}$

$$(I_0(\alpha))^2 = \int_{-\infty}^{\infty} du \int_{-\infty}^{\infty} dv \; e^{-\frac{\alpha}{2}(u^2+v^2)} = \int_0^{\infty} 2\pi \, r dr \; e^{-\frac{\alpha}{2} r^2} =$$

$$= \pi \int_0^{\infty} d(r^2) \; e^{-\frac{\alpha}{2} r^2} = \left(-\frac{2\pi}{\alpha}\right) e^{-\frac{\alpha}{2} r^2} \Big|_0^{\infty} = \frac{2\pi}{\alpha}$$

provided that $e^{-\frac{\alpha}{2} r^2} \Big|_{r=\infty} = 0$

which means $Re(\alpha) > 0$.

Next consider $I_1(\alpha, j) = \int_{-\infty}^{\infty} du \; e^{-\frac{\alpha}{2} u^2 + ju} =$

$$= \int du \; e^{-\frac{\alpha}{2}\left(u^2 - \frac{2j}{\alpha} u\right)} = \int_{-\infty}^{\infty} du \; e^{-\frac{\alpha}{2}\left[\left(u - \frac{j}{\alpha}\right)^2 - \frac{j^2}{\alpha^2}\right]} =$$

$$= \left(v = u - \frac{j}{\alpha}\right) = \int_{-\infty}^{\infty} dv \; e^{-\frac{\alpha}{2} v^2 + \frac{j^2}{2\alpha}} =$$

$$= \sqrt{\frac{2\pi}{\alpha}} \; e^{\frac{1}{2\alpha} j^2}$$

Another way: $\int_{-\infty}^{\infty} dx \frac{d}{dx}\left(e^{-\frac{\alpha}{2} x^2 + jx}\right) = 0$ (for $Re(\alpha) > 0$)

But this means $\int_{-\infty}^{\infty} dx \left(-\alpha x + j\right) e^{-\frac{\alpha}{2} x^2 + jx} = 0$

$$\Rightarrow \; -\alpha \int_{-\infty}^{\infty} dx \underbrace{e^{-\frac{\alpha}{2} x^2 + jx}}_{x} + j \underbrace{\int_{-\infty}^{\infty} dx \; e^{-\frac{\alpha}{2} x^2 + jx}}_{I_1(\alpha, j)} = 0$$

Notice that $\int dx \; x e^{-\frac{\alpha}{2} x^2 + jx} =$

$$= \frac{\partial}{\partial j} \int dx \; e^{-\frac{\alpha}{2} x^2 + jx} = \frac{\partial I_1(\alpha, j)}{\partial j}$$

$\Rightarrow$ we obtain dif. equation for $I_1(\alpha, j)$:

$$-\alpha \frac{\partial I_1}{\partial j} + j I_1 = 0$$

$$\frac{\partial I_1}{\partial j} = \frac{j}{\alpha} I_1 \Rightarrow I_1(\alpha, j) = C e^{\frac{1}{2\alpha} j^2}$$

Initial condition: $I_1(\alpha, j=0) = I_0(\alpha) = \sqrt{\frac{2\pi}{\alpha}}$

$$\Rightarrow I_1(\alpha, j) = \sqrt{\frac{2\pi}{\alpha}} e^{\frac{j^2}{2\alpha}} \quad \text{(as before)}$$

2) Consider now Fresnel integral

$$J_0(a) = \int_{-\infty}^{\infty} dx\, e^{\frac{ia}{2}x^2} = I_0(-ia)$$

Observe that $I_0(\alpha) = \sqrt{\frac{2\pi}{\alpha}}$ as long as $Re(\alpha) > 0$

Write $\alpha = \underset{Re(\alpha)}{\alpha'} + i\underset{Im(\alpha)}{\alpha''} = -ia = -i(a' + ia'') =$

$= -ia' + a''$

$\Rightarrow$ condition $Re(\alpha) > 0$ corresponds to $Im(a) = a'' > 0$

$$\Rightarrow J_0(a) = I_0(-ia) = \sqrt{\frac{2\pi}{-ia}} = \sqrt{\frac{2\pi i}{a}} = \sqrt{\frac{2\pi}{a}} e^{i\pi/4}$$

as long as $Im(a) > 0$ (can be <u>infinitesimal</u>)

Next $J_1(a, J) = \int_{-\infty}^{\infty} dx\, e^{i\frac{a}{2}x^2 + iJx} = I_1(-ia, iJ) =$

$$= \sqrt{\frac{2\pi}{-ia}}\, e^{\frac{(iJ)^2}{2(-ia)}} = \sqrt{\frac{2\pi i}{a}}\, e^{-\frac{i}{2a}J^2}$$

which can also be checked by completing the square in $e^{\frac{i}{2}ax^2 + iJx}$.

Matrix generalization of Gaussian integrals:

3) $$\int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy\, e^{-\frac{\alpha}{2}x^2 - \frac{\beta}{2}y^2} = \sqrt{\frac{2\pi}{\alpha}}\sqrt{\frac{2\pi}{\beta}} = \frac{2\pi}{\sqrt{\alpha\beta}}$$

Moreover $K_2 = \int dx \int dy\; e^{-\frac{1}{2}(\alpha x^2 + \beta y^2 + 2\gamma xy)} = \dfrac{2\pi}{\sqrt{\alpha\beta - \gamma^2}}$

because $K_2 = \int dx\, dy\; \exp -\frac{1}{2}\left[\alpha\left(x + \frac{\gamma}{\alpha}y\right)^2 - \alpha\left(\frac{\gamma y}{\alpha}\right)^2 + \beta y^2\right]$

$$= \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dx\; \exp\left\{-\frac{\alpha}{2}\left(x + \frac{\gamma}{\alpha}y\right)^2 - \frac{1}{2}\left(\beta - \frac{\gamma^2}{\alpha}\right)y^2\right\}$$

let $\begin{cases} u = x + \frac{\gamma}{\alpha}y \\ v = y \end{cases}$

$$\Rightarrow \int du\, dv_{m} = \int J\, dx\, dy_{m}, \quad J = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & \frac{\gamma}{\alpha} \\ 0 & 1 \end{vmatrix} = 1$$

$$\Rightarrow K_2 = \int_{-\infty}^{\infty} dv \int_{-\infty}^{\infty} du\; e^{-\frac{\alpha}{2}u^2 - \frac{1}{2}\left(\beta - \frac{\gamma^2}{\alpha}\right)v^2} =$$

$$= \sqrt{\frac{2\pi}{\alpha}} \cdot \sqrt{\frac{2\pi}{\beta - \frac{\gamma^2}{\alpha}}} = \frac{2\pi}{\sqrt{\alpha\beta - \gamma^2}}$$

Now observe that $\alpha\beta - \gamma^2 = \det(A)$

where $\alpha x^2 + \beta y^2 + 2\gamma xy = (x, y)\, A \begin{pmatrix} x \\ y \end{pmatrix} =$

$$= (x, y)\begin{pmatrix} \alpha & \gamma \\ \gamma & \beta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow K_2 = \frac{2\pi}{(\det A)^{1/2}} = \int dx\, dy\; e^{-\frac{1}{2}(x,y)\, A \begin{pmatrix} x \\ y \end{pmatrix}}$$

$$\Rightarrow K_N = \int dx_1 dx_2 \ldots dx_{N-1}\; e^{-\frac{1}{2}\vec{z}^T A_N \vec{z}} = \frac{(2\pi)^{N/2}}{\sqrt{\det A_N}}$$